Hibernation

In the fall, many insect-eating birds migrate to warmer climes, leaving the forest's remaining animals to face the winter cold. Animals that store food can remain active throughout this difficult time of year, but others use a very different survival strategy: they hibernate, living on the fat reserves they have built up during the summer months.

How long and how deeply an animal hibernates depends on where it lives. In the forests of northwest Europe, hedgehogs may hibernate for up to 6 months, whereas farther south, their winter sleep is much shorter. In eastern North America, woodchucks—or groundhogs—typically hibernate from October to February; their wanderings early in the year are a traditional sign that spring is not far off.

Some hibernating animals, such as the common dormouse, hardly ever interrupt their winter break, even if they are picked

SLEEPING THROUGH THE COLD
Common dormice use leaves and moss to make well-insulated winter nests among brambles and other plants. Here, part of the nest has been temporarily removed to reveal the hibernating animal, fast asleep with its tail wrapped tightly around its body.

up. However, many hibernators behave in a different way. If the weather turns warm, they briefly rouse themselves: bats, for example, take to the air for feeding flights, while hedgehogs often move out of one hibernation nest and into another. But forest hibernators have to be careful not to do this too often: activity uses up their bodily food reserves, and it therefore puts them at risk of running out before the winter is truly over. Many insects also hibernate, often hidden under bark; but in some species, the adults die out, leaving behind tough, overwintering eggs that will hatch in spring.

NIGHT FLIGHT
Unlike most squirrels, North American flying squirrels feed at night. The elastic skin flaps between their legs enable them to glide for over 165 ft (50 m). They steer with their front legs, and use their large eyes to navigate safely in the dark.

Movement

While monkeys and gibbons are the most impressive climbers in tropical forest, squirrels are the experts in temperate forest. Unlike many climbing mammals, they can run head-first down tree trunks, as well as up them, by hooking their long, curved hind claws into bark. Squirrels have excellent eyesight, and they instinctively scuttle to the back of a tree if they spot a potential predator—a simple behavior that makes them difficult to catch.

Temperate forest is inhabited by gliding rodents, and also—in Australasia—by gliding marsupials. But for precise maneuvering among trees, owls and birds of prey are unrivaled. Unlike their relatives in open habitats, most of these aerial hunters have relatively short, broad wings that enable them to twist and turn effectively. A prime example of this adaptation to woodland life is the Eurasian sparrowhawk: rather than soaring and then swooping, it speeds among trees and along hedgerows—sometimes only a yard or so above ground—ambushing small birds in midair and carrying them away in its talons.

Temperate forest provides ground-dwelling animals with lots of cover. As a result, small mammals, such as voles and shrews, abound on the forest floor. To avoid being seen as they move around, these animals often use runways partly covered by grass or fallen leaves. Voles use a combination of vision, smell, and touch to find their way along these runs, but shrews, which have very poor eyesight, navigate partly by emitting high-pitched pulses of sound. These signals bounce back from nearby objects in the same way as those sent out by a bat's sonar system.

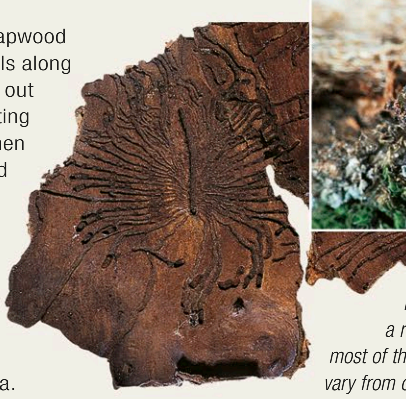
Living in leaf litter

The leaf litter in temperate forest is one of the world's richest animal micro-habitats. This deep layer of decomposing matter harbors vertebrates, including mammals and salamanders, but its principal inhabitants are invertebrates that feed on leaf fragments, on fungi and bacteria, or on each other. Some of these animals—such as centipedes and woodlice—are large enough to be easily seen, but many others are microscopic. Animals that live deep in leaf litter exist in total darkness, so most of them rely on their sense of touch to find food. This is especially true of predators: centipedes locate their prey with long antennae, while tiny pseudoscorpions use the sensory hairs that cover their pincers. Like true scorpions, pseudoscorpions are venomous; but they are so small that they pose no threat to anything much bigger than themselves. This is fortunate because in just a few square yards of leaf litter their numbers can run into millions. Dead leaves are a useful screen, hiding leaf-litter dwellers from other animals foraging on the forest floor. However, it is not totally secure. Some temperate forest birds, particularly thrushes, pick up leaves and toss them aside, snapping up leaf-litter animals as they try to rush away from the light.

HUNTER IN THE LEAVES
Lithobiid centipedes, such as this one, have flattened bodies that allow them to crawl under leaves and fallen logs as well as on the ground surface. Geophilid centipedes live permanently underground, and therefore have narrow, almost wormlike bodies.

INSECT ENGRAVERS

Female bark beetles tunnel through the sapwood beneath living bark, laying eggs at intervals along the way. The hatched larvae eat their way out at right angles to the original tunnel, creating distinctive "galleries" that can be seen when dead bark falls away. The side tunnels end in exit holes, from which the developed beetle emerges and flies away. Common in deciduous forest, bark beetles can be highly destructive because they often infect trees with fungi. One species—the elm bark beetle—spreads Dutch elm disease, a fungus that has wiped out elms in parts of Europe and North America.



BARK GALLERIES
The adult bark beetle has a cylindrical body and a round-fronted thorax hiding most of the head. The gallery patterns vary from one species to another.